a

60 studies
23,682 experimental records
21,197 unique LNPs
14,993 ionizable lipids
60 publications
40 biological models
75 biological targets
7 cargo types
3 experimental settings
8 outcome families
99 retained fields

b

LNPhub_YX_2024
LNPhub_KZ_2016
LNPhub_LM_2019
LNPhub_YZ_2022
LNPhub_LX_2024
LNPhub_JL_2024
LNPhub_SL_2021
LNPhub_AJ_2023
LNPhub_BL_2023b
LNPhub_QL_2026
LNPhub_ZH_2023
LNPhub_CW_2025
LNPhub_YR_2024
LNPhub_LX_2024b
LNPhub_XG_2025
LNPhub_YX_2023
LNPhub_SW_2024
LNPhub_XH_2024c
LNPhub_XH_2024b
LNPhub_MP_2025
LNPhub_SX_2025
LNPhub_ZC_2023
LNPhub_JL_2023
LNPhub_YW_2025
LNPhub_AP_2025
LNPhub_DZ_2021
LNPhub_LL_2026
LNPhub_SY_2025
LNPhub_SB_2024
LNPhub_SS_2018

Ionizable lipid structure
Formulation
Physicochemical
Delivery
Viability
Toxicity
Editing
Uptake
Immune
Protein / other outcomes

% records with value
100
80
60
40
20
0

c

Experimental setting
in vitro 20,015
in vivo 3,576
Not reported 54
ex vivo 37
0 10,000 20,000
Records

Cargo type
mRNA 19,002
siRNA 3,355
pDNA 1,093
mRNA/sgRNA 73
Not reported 65
siRNA/mRNA 39
0 5,000 10,000 15,000
Records

d

Biological model
HeLa 7,873
A549 2,526
HepG2 2,255
DC2.4 1,843
Mouse C57BL/6 1,814
Other 1,729
RAW264.7 1,254
IGROV1 1,206
BEAS-2B 1,022
Mouse BALBc 751
Mouse C57BL/6J 744
0 2,000 4,000 6,000 8,000
Records

Biological target
lung epithelium 1,836
bronchial epithelium 1,074
Other 1,060
muscle 1,003
liver 942
multiorgan 331
spleen 286
colon cancer cells 269
lung 152
dendritic cells 140
bone marrow 139
0 250 500 750 1,000 1,250 1,500 1,750
Records

Administration route
intravenous 2,295
intramuscular 1,025
topical transdermal 83
intratracheal 66
intradermal 66
inhalation 38
subcutaneous 3
0 500 1,000 1,500 2,000
In vivo records

e

UMAP 2
UMAP 1
biodegradable amino lipid
amino lipid
Other
phospholipid
biodegradable branched lipid
biodegradable lipid
biodegradable lipidoid
heterocyclic lipid
biodegradable dendrimeric lipid

f

reaction type
multi-step/other
multi-step dendrimer synthesis
Michael addition
Ugi-4CR
Ugi-3CR

il architecture type
head–tailA–tailB
head–linker–tail
head–linker–two-tail
head–linker–multi-tail

il tail count
Not reported
2.0
1.0

g

delivery
base editing percent
editing relative fvii ec50 mg per kg
editing gfp negative cells percent
immune il6 pg ml
immune tnf alpha pg ml
immune ifn gamma pg ml
immune il1a pg ml
immune gm csf pg ml
protein abundance value
gfp mfi transfection
mrna binding efficiency percent
particle size nm
pdi
zeta potential mv
encapsulation efficiency percent
measured pka
toxicity hemolysis percent
toxicity alt u per l
toxicity ast u per l
toxicity cera umol per l
toxicity urea umol per l
toxicity bun mg per dl
cellular metabolic uptake percent
uptake cy5 mluc lnps percent
viability igrov1 percent
viability mtt 5 ug per ml
viability mtt 0 5 ug per ml
10^1 10^2 10^3 10^4
Non-null measurements